Computational Campaigns for Soft and Hard FTL: Baselines, Closeouts, and Prototype Readiness

Technology Program Draft

Draft skeleton

Abstract

This manuscript is the canonical computational-paper skeleton for the faster-than-light transport program. It normalizes the frozen baseline, soft-array closeout, hard-bench closeout, and remaining mission-level stages into one reproducible narrative that will later be backed by a live canonical source tree.

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1 Scope and Numerical Status

The FTL computational line consists of staged simulators, closeout packages, and integrated policy layers. This paper will present them as computational evidence and engineering readiness artifacts, not as standalone proof of the theory.

2 Frozen Package Ledger

This chapter will normalize the package hierarchy from abstract simulators to medium-specific packages, integrated mission stacks, and stage closeouts.

3 Baseline Freeze

This chapter will present the canonical frozen baseline package and explain how the baseline metrics, regression checks, and freeze summaries define the reference computational state.

4 Soft Numerical Closeout

This chapter will describe the progression from the abstract soft witness line to the medium-specific array closeout package.

4.1 Soft witness and array metrics

Placeholder for success rate, margins, repeatability, and failure fractions.

4.2 Soft closeout status

Placeholder for the completed soft S1-M1 array closeout stage.

5 Hard Numerical Closeout

This chapter will describe the progression from hard one-pole simulators to the H1-M2 bench closeout package.

5.1 One-pole and composite metrics

Placeholder for fidelities, binding margins, topological protection, and cycle repeatability.

5.2 Hard closeout status

Placeholder for the completed hard H1-M2 bench closeout stage.

6 Mission, Portfolio, and Adaptive Policy Layers

This chapter will normalize the orchestration, mission scenario, portfolio, optimizer, and adaptive-policy packages as one integrated control stack.

7 Pending Closeout Stages

This chapter will explicitly mark the integrated mission campaign, medium downselection, and prototype dossiers as pending clean outputs despite their partial legacy presence.

8 Bridge to Physical Prototype Protocols

This chapter will explain how the numerical closeouts feed soft-array protocols, hard-bench protocols, and integrated soft-to-hard scheduling.

9 Reproducibility Contract

This chapter will define the canonical source-tree target, config normalization, regression requirements, and report-generation rules that the later code import must satisfy.

10 Limitations and Non-Claims

This chapter will separate numerical readiness from physical demonstration and will restate the observer-relative and medium-dependent scope limits.

11 Forward Dependencies

This chapter will list the live-source import tasks, test requirements, and Lean-adjacent definitions that depend on the computational normalization.